PRESENTED BY

GENERATIVE GAINS

# Thermal Trace

PRE-SEED · QAR 1.85M · QATAR FIRST, GCC NEXT

## Know what a failing fridge is costing before the gate does.

Cold-chain intelligence: physical risk and commercial value, in one reserve.



• CITED · FAO / UNEP 2024

# A third of our food never gets eaten.

12.3 SDG 12.3 · halve food waste, cut supply-chain loss

2 SDG 2 · zero hunger

13 SDG 13 · climate action

**13%**

of food is **lost** between harvest and retail.

**19%**

more is **wasted** after it reaches retail and homes.

**8–10%**

of global greenhouse-gas emissions come from food loss and waste.

● CITED · TASMU · QATAR NFSS 2030

# An import-dependent region, a brutal summer.

## Climate and imports



**45 °C**

Imports by sea and land, customs holds and border queues, across a summer that passes 45 °C. A fridge that fails at noon can spoil a load in hours.

## Policy tailwind

TASMU analytics and Qatar's Food Security Strategy target efficient supply chains and less food waste. Thermal Trace works on the loss in transit.

NATIONAL FOOD SECURITY STRATEGY 2030

| | | |
|---|---|---|
| Food waste | | −50% |
| Food loss | | −30% |

THERMAL DEBT — Maritime import + Customs hold + Border queue + 45 °C summer = Lost shelf life

Thermal Trace is the **operator-level layer** that complements national systems.

○ COMMERCIAL RESERVE · DESIGNED, STAGED IN THE PILOT

# Safe is not the same as sellable.

| Thermal Reserve<br>THE PHYSICAL CLOCK | Commercial Reserve<br>THE COMMERCIAL CLOCK |
|---|---|
| FOCUS<br>Product health: “Can it still be consumed?” | FOCUS<br>Economic life: “Can it still earn its value?” |
| CLOCK<br>Production → Transport → Deterioration → Shelf-life boundary | CLOCK<br>Arrival → Customer exposure → Sell-through → Replenishment |
| GOAL<br>Protect physical life. ● CALCULATED | GOAL<br>Protect economic life. ● PREDICTED |

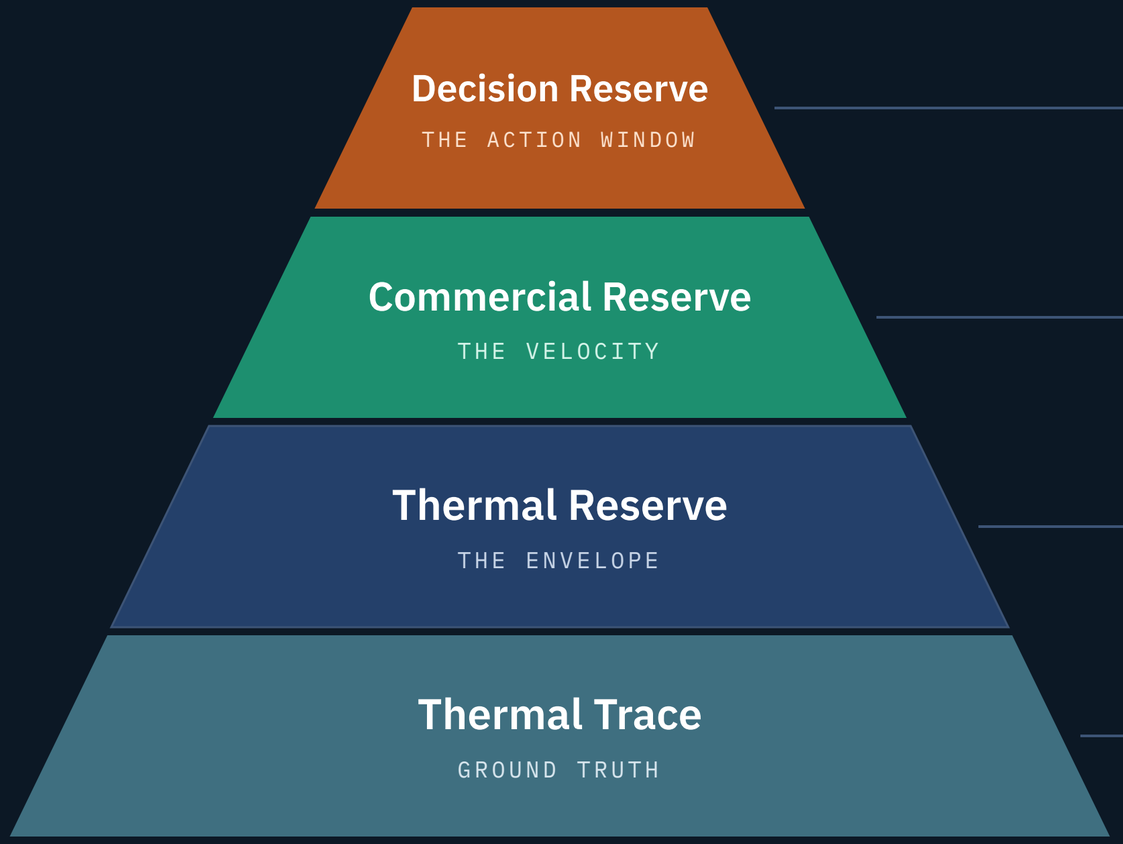
Health creates the possibility of selling. Demand creates the possibility of realising value.
**Commercial Reserve lives where the two meet.**

$$CR = Q_r \times P_r \times V_r$$

$Q_r$ quantity realistically realisable
$P_r$ probability of sale
$V_r$ expected value per unit

○ LAYER 3 DESIGNED · STAGED IN THE PILOT

# The anatomy of value: four reserves, one stack.




LAYER 4 · DECISION RESERVE ● OPTIMIZED

The time left to act before the good options disappear.

LAYER 3 · COMMERCIAL RESERVE ○ DESIGNED

The economic runway: value that can still be realised.

$$CR = Q_r \times P_r \times V_r$$

LAYER 2 · THERMAL RESERVE ● CALCULATED

The physical budget left before quality is compromised.

LAYER 1 · THERMAL TRACE ● MEASURED

A continuous, timestamped memory of temperature exposure and stress.

# Six signals. Three numbers that matter.

● TEMPERATURE

**DHT11 today · DS18B20 next**

The demo node reads one point. Ceiling and floor probes are the next step, so a warm layer shows up instead of hiding in an average.

● HUMIDITY

**DHT11 today · SHT3x next**

Rising humidity shows whether an open door is wetting the cartons.

○ SHOCK & TILT · SIMULATED

**3-axis MEMS · designed**

Bruising from a dropped pallet is damage a thermometer never sees.

○ GASES · DESIGNED

**Metal-oxide · VOC, $C_2H_4$**

Ethylene is the fruit itself reporting that it has started to ripen.

○ SUB-SURFACE · DESIGNED

**NIR · 700–2500 nm**

Finds internal bruising before anything is visible from outside.

○ SURFACE · DESIGNED

**RGB camera · edge vision**

Colour shift, mould, and a count of what is actually on the shelf.

THEY FEED → Deterioration detection · Remaining Useful Time · Thermal exposure $E_T$ ● CALCULATED

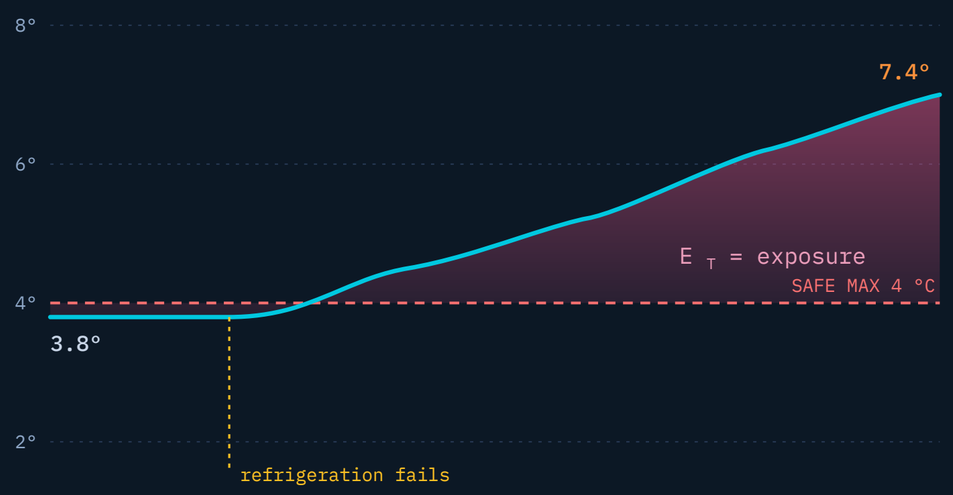
Temperature and humidity flow from a real sensor node today (DHT11, about ±2 °C); shock is simulated and the rest are the designed next layer. All publish the same message, so adding one changes nothing downstream.

○ EXAMPLE SCENARIO

# The label says 7 days. The trace says 4.5.

LAYER 1 · THERMAL TRACE · CARGO TEMPERATURE

● MEASURED




THERMAL EXPOSURE · °C·MIN

$$E_T = \Sigma \max(0, T_i - T_{safe}) \cdot \Delta t$$

DETERIORATION RATE · ARRHENIUS

$$k(T) = k_{ref} \cdot e^{(E_a/R)(1/T_{ideal} - 1/T)}$$

LAYER 2 · THERMAL RESERVE · SAFE LIFE LEFT

$$D = \Sigma\, k(T_i)\cdot\Delta t_i \rightarrow \text{reserve} = L\cdot(1 - D)$$

| | | |
|---|---|---|
| LABEL DATE | | 7.0 d |
| THERMAL RESERVE | | 4.5 d |

● CALCULATED Deterministic Python. No AI model touches these numbers.

AI ON A LEASH
NEITHER MAY INVENT
A NUMBER

**Laya** SYSTEM 1 · LOCAL · OFFLINE ● PREDICTED

Answers typed questions in one pass. Never generates text; corroborates the decision, never overrides it. Base model, confidence shown as uncalibrated.

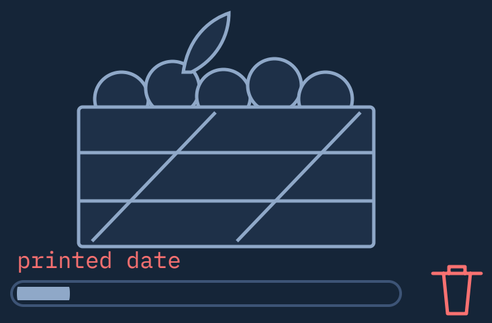
**DeepSeek V4.1 Flash** SYSTEM 2 · CLOUD ● AI-EXPLAINED

Explains the calculated facts in plain words. May rank only feasible options; never invents a temperature, ETA or cost.

# Redefining the perishable asset.

## The biological clock

DEFENSIVE · TODAY

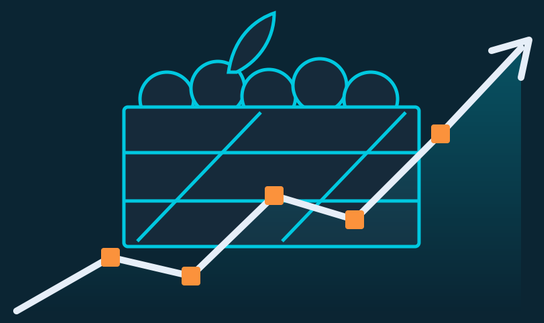



**Legacy model:** inventory is a degrading liability, managed by a static printed expiry date.

Result: late markdowns and write-offs.

## The commercial trace

OFFENSIVE · THERMAL TRACE



**Thermal Trace model:** inventory is an asset with a live value, priced from its real thermal history and live demand.

Result: value recovered before it is lost.

○ COUNTDOWN · EXAMPLE · COMMERCIAL RESERVE DESIGNED

# Two clocks, one platform.

**Thermal Trace** tells us what is happening to the product. Two reserves read it.

PHYSICAL CLOCK

● CALCULATED

## Thermal Reserve

How much physical margin remains.

**45** min left

For 3PLs, fleets and warehouses. Sold as a platform fine-tuned to each customer: sensors, calibration and integration.

3PLs | Fleets | Warehouses

COMMERCIAL CLOCK

● PREDICTED

## Commercial Reserve

How much economic opportunity remains, and what we should do about it.

POS · ERP · WMS → route → price to clear

For hypermarkets and HORECA. Sold as SaaS: joins the physical state with POS, ERP and WMS to route stock, and price it to clear.

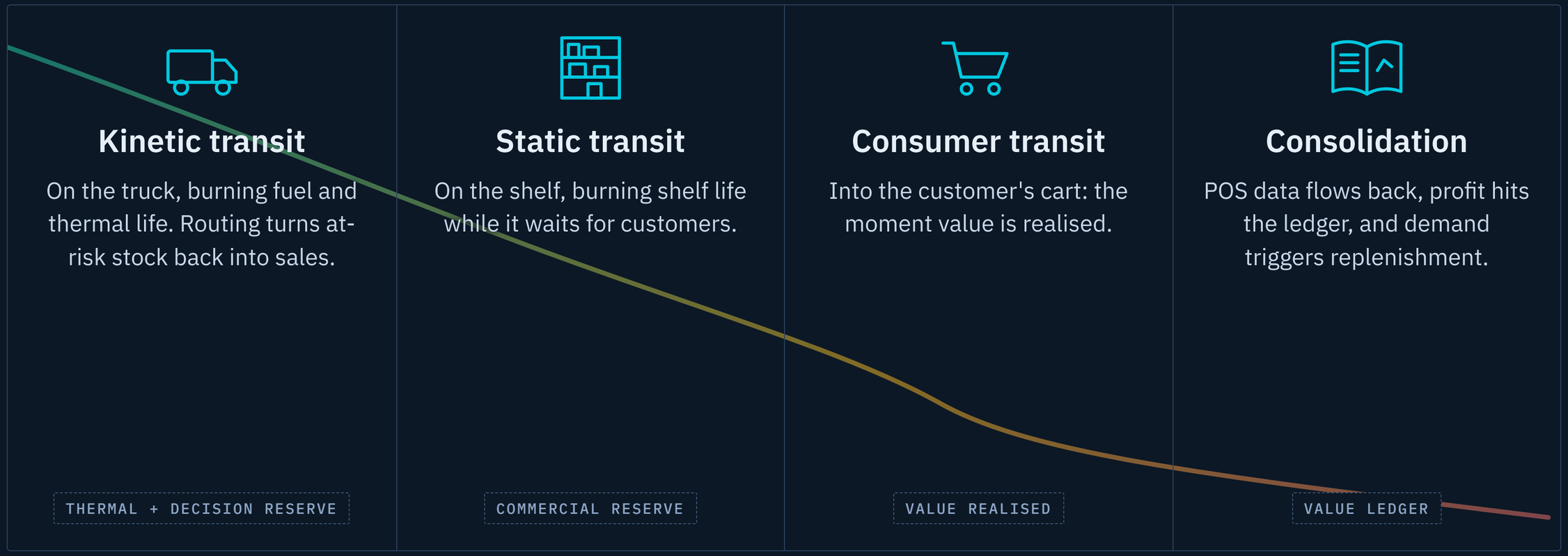
Hypermarkets | HORECA

ENTRY POINT · THERMAL RESERVE · TUNED PLATFORM ——→ CONTROLLED EXTENSION · COMMERCIAL RESERVE · SAAS

ILLUSTRATIVE VALUE CURVE · DESIGNED

# In transit until the profit is booked.

The trace follows the value from farm to ledger, not just from dock to dock.



### Kinetic transit

On the truck, burning fuel and thermal life. Routing turns at-risk stock back into sales.

THERMAL + DECISION RESERVE

### Static transit

On the shelf, burning shelf life while it waits for customers.

COMMERCIAL RESERVE

### Consumer transit

Into the customer's cart: the moment value is realised.

VALUE REALISED

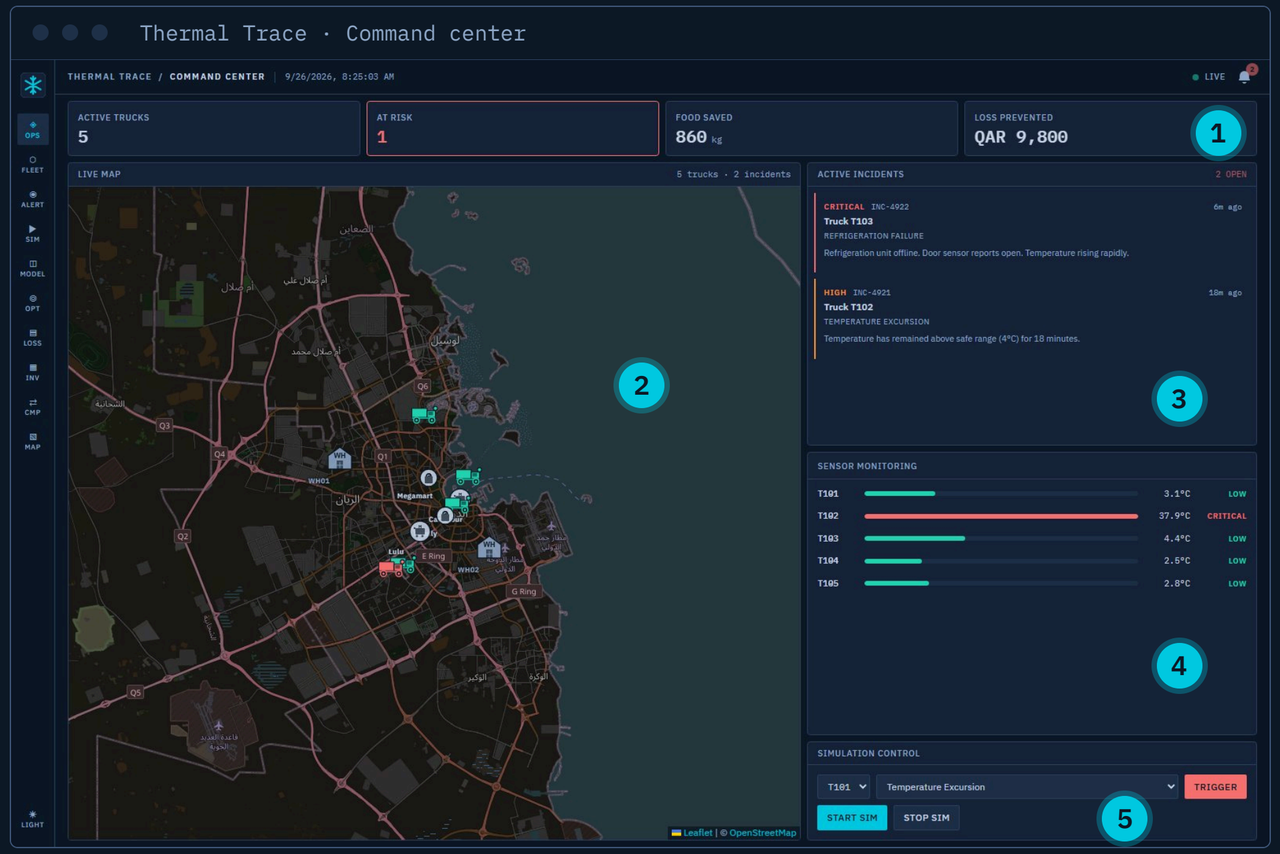
### Consolidation

POS data flows back, profit hits the ledger, and demand triggers replenishment.

VALUE LEDGER

○ LIVE DEMO · SIMULATED DATA

# One screen for the whole cold chain.




1. **Headline figures**
   Active trucks, at risk, food saved, loss prevented.
2. **Live map**
   Every truck, warehouse and store, coloured by risk.
3. **Active incidents**
   Ranked by severity, with the cause in plain words.
4. **Sensor monitoring**
   Each truck's cargo temperature against its safe range.
5. **Simulation control**
   Trigger a failure and watch the system respond.

Data → Condition → Prediction → Optimization → Decision → Food loss

○ EXAMPLE SCENARIO

# The window to act, and the move to make.

$$\min( C_{transport} + C_{food\text{-}loss} + C_{delay} )$$

SUBJECT TO

- ETA ≤ remaining thermal reserve
- Quantity ≤ available capacity
- Storage temperature compatible

AI — Corroborates — Explains — Never invents a number

● REEFER Compressor failing — **Thermal reserve 45 min**

| DESTINATION | ETA | FITS 45 MIN |
|---|---|---|
| **Hypermarket** Planned route | 90 min | ✗ spoils |
| **Nearby buyer** Re-routed | 20 min | ✓ fits |

● OPTIMIZED **REROUTE**

Decision reserve ≈ 25 min to commit. The optimizer picks; the AI only explains why.

ILLUSTRATIVE · PRODUCTS NAMED FOR REFERENCE

# Banana fails cold. Ice cream fails warm.

| SIGNAL | 🍌 13–14 °C · 90–95 % RH<br>Too cold browns it, ethylene ripens it | 🍦 −18 °C or colder<br>Any thaw-refreeze ruins texture | Gate check only<br>One logger, read after delivery | Bought trackers<br>In transit vs handheld at QA |
|---|---|---|---|---|
| TEMPERATURE<br>DHT11 today · DS18B20 next | Below 13 °C = chilling injury: peel turns grey days later, after it is sold on. | A rise above −18 °C grows ice crystals; it refreezes gritty and nobody can tell from outside. | Excursion found at the gate or by the customer. Load already damaged. | Tive Solo 5G, Sensitech TempTale GEO X, ELPRO LIBERO |
| HUMIDITY<br>DHT11 today · SHT3x next | Too dry and skins shrivel and scar; too wet and cartons go soft. | A door left open lets moist air in: frost on cartons proves a warm-up. | Soggy or dry cartons noticed only when unloading. | Tive Solo 5G, ELPRO LIBERO Gx |
| SHOCK & TILT · SIMULATED<br>3-axis MEMS · designed | A dropped pallet bruises; bruises appear 1–2 days later. | Crushed tubs and popped lids; tilt lets softened product leak. | Damage claims argued with no record of where or when. | Tive Solo 5G (12 g shock), Controlant Saga (vibration) |
| GASES · DESIGNED<br>Metal-oxide · VOC, $C_2H_4$ | Ethylene rising: the load started ripening in transit, so sell it first. | *Not used. Sealed product, and MOx sensors are not rated this cold.* | Over-ripe load discovered at the store, too late to reroute. | *Handheld only: Felix F-900 ethylene analyser* |
| SUB-SURFACE · DESIGNED<br>NIR · 700–2500 nm | Starch turning to sugar inside, before the peel shows it. | *Not used. Packaging blocks it; temperature history tells the story.* | Ripeness judged by eye at the dock. | *Handheld only: Felix F-750, VIAVI MicroNIR* |
| SURFACE · DESIGNED<br>RGB camera · edge vision | Peel colour stage, brown spots, crown mould, count on shelf. | Frosted or misshapen packs, open lids, empty facings. | Staff walk the aisle a few times a day. | *No in-transit product; fixed shelf cameras only* |

THEY FEED → Deterioration detection · Remaining Useful Time · Thermal exposure $E_T$ · CALCULATED

# From managing biology to managing liquidity.

| DIMENSION | LEGACY SUPPLY CHAIN | THERMAL TRACE |
|---|---|---|
| THE GOAL | Delay spoilage (defensive) | Maximise yield (offensive) |
| ASSET VIEW | A degrading physical object | An asset with a live, priced value |
| INVENTORY RULE | FEFO: first expired, first out, by printed date | LCRFO (planned): lowest Commercial Reserve, first out, by live physics |
| INTERVENTION | Passive temperature alarms | Optimized routing today; early markdowns planned |
| WASTE | An invisible operating tax | A quantified, bounded value at risk |

HYPOTHESES TO VALIDATE · NOT MARKET PRICES

# Tailored to each customer. Sold as SaaS.

THERMAL TRACE PLATFORM · SHARED IP

Trace engine | Food-state model | Data ingestion | Deterministic governance | Decision workflows | Value ledger

PHYSICAL INTELLIGENCE

## Thermal Reserve

Configured to the customer's equipment, sensors and products. The deployment is customised; the platform stays reusable.

MARKET INTELLIGENCE

## Commercial Reserve

Precise calculations over the retailer's own POS, inventory and ERP data. The repeatable SaaS product.

**Pay 1×, recover ≥ 3×** every QAR 1 charged saves QAR 3+ of food.

1 · ONE-TIME SETUP

### Implementation — QAR 90K

Fixed setup cost. Hardware they need is billed separately, at cost.

→

2 · YEARLY SUBSCRIPTION · PICK ONE

### Standard — QAR 120K / yr

One product · live monitoring, alerts and reserves · monthly value report · email support

2 · YEARLY SUBSCRIPTION · PICK ONE

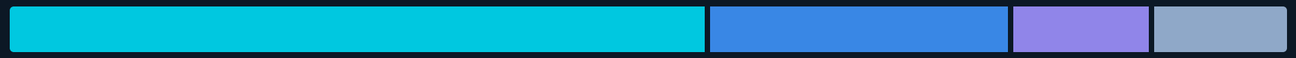
### Enterprise — QAR 216K / yr

Both products · customised dashboard · unlimited integrations · SSO and uptime SLA · full support

# Twelve months to prove it in the field.

## QAR 1,850,000

≈ US$508,000 · 12 months · pre-seed or strategic round



| | |
|---|---|
| Core five-person engineering team | 55.1% |
| Sensors, deployment, cloud, integration | 23.5% |
| Validation, customers, legal | 10.8% |
| Contingency reserve | 10.5% |

### WHAT IT BUYS IN 12 MONTHS

01 An anchor customer live at 3–5 facilities

02 Avoided loss measured at ≥ 3× the platform fee

03 Commercial Reserve tested in controlled trials

04 An annual contract and a second customer

05 QAR 300–500K contracted ARR TARGET

12.3 Every kilogram saved is logged as evidence.

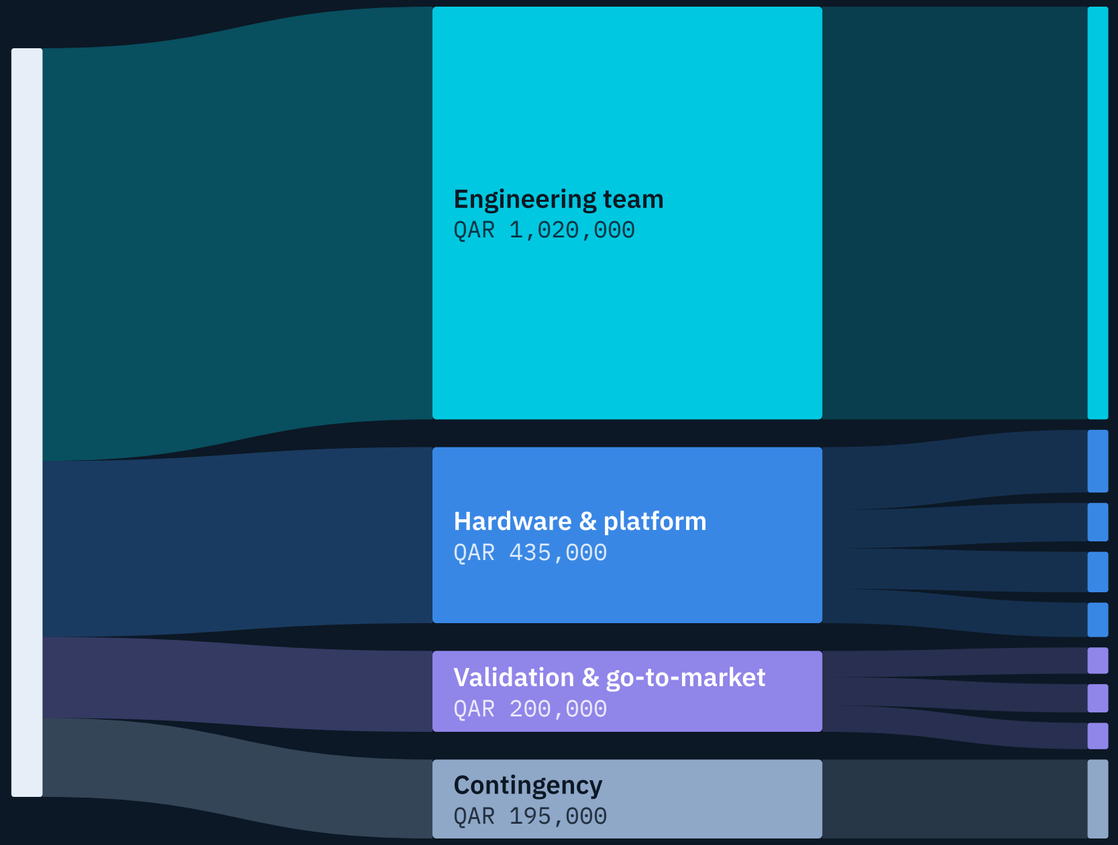
PRE-SEED BUDGET · QAR

# Every riyal, from round to line item.

PRE-SEED ROUND

**QAR 1.85M**

≈ US$508K · 12 mo




| | | |
|---|---|---|
| Core five-person team | **1,020,000** | 55.1% |
| Sensors, edge equipment | **155,000** | 8.4% |
| Deployment, calibration | **95,000** | 5.1% |
| Cloud, data, security | **100,000** | 5.4% |
| Enterprise integration | **85,000** | 4.6% |
| Food-domain validation | **65,000** | 3.5% |
| Customer development | **70,000** | 3.8% |
| Legal, IP, readiness | **65,000** | 3.5% |
| Contingency reserve | **195,000** | 10.5% |

○ PLANNING ASSUMPTIONS · NOT MARKET RATES

# Five engineers, QAR 76,000 a month.

| ROLE | OWNS | MONTHLY (QAR) | ANNUAL (QAR) |
|---|---|---|---|
| **CTO · Systems & Product Architect** | Architecture, scope, data contracts, security, delivery | 20,000 | 240,000 |
| **Thermal Intelligence · Data Engineer** | Thermal state, exposure, risk models, shelf-life logic | 16,000 | 192,000 |
| **Backend & Data Platform Engineer** | Ingestion, databases, APIs, identity, integrations | 15,000 | 180,000 |
| **IoT & Edge Engineer** | Sensors, gateways, connectivity, calibration | 13,000 | 156,000 |
| **Application & Workflow Engineer** | Operator UI, alerts, approvals, ERP and WMS workflows | 12,000 | 144,000 |
| Total gross compensation | | 76,000 | 912,000 |
| Employment-cost provision | | | ≈ 109,000 |
| **Total team budget** | | | **≈ 1,020,000** |

The CEO leads investors, anchor customers and sales. Food-safety experts are hired fractionally, from the validation line.

# The other QAR 830K, line by line.

| | | |
|---|---|---|
| **HARDWARE & PLATFORM** | | **435,000** |
| Sensors and edge equipment | Sensors and gateways for 3–5 facilities: cold rooms, reefer trucks, dispatch areas | 155,000 |
| Deployment, calibration, field testing | Install, calibrate, and test outages and alerts on site | 95,000 |
| Cloud, data infrastructure, security | 12 months of hosting, storage and security for asset history | 100,000 |
| Enterprise integration, data engineering | A few high-value flows into product, shipment, ERP and WMS records | 85,000 |
| **VALIDATION & GO-TO-MARKET** | | **200,000** |
| Thermal and food-domain validation | Fractional food-safety experts review thresholds and shelf-life models | 65,000 |
| Customer development and success | The anchor pilot: site survey, value baseline, conversion | 70,000 |
| Legal, IP, commercial readiness | Data-access agreements, pilot and annual contracts, IP | 65,000 |
| **CONTINGENCY RESERVE** | released only with documented founder and investor approval | **195,000** |

# From pilot to recurring revenue.

### Months 1–4

Secure the pilot, agree how value is measured, install the devices.

### Months 5–9

Run the intervention workflow and prove the avoided loss against the baseline.

### Months 10–12

Convert to an annual contract and win a second customer.

The pilot must prove avoided loss **≥ 3× the platform fee**. Month-12 target: **QAR 300–500K contracted ARR**. A target, not a forecast.

# Known risks, planned responses.

| RISK | MITIGATION |
| --- | --- |
| Customer data access | Sign data access and a named sponsor first |
| False precision | Validated assumptions, uncertainty shown, domain review |
| Integration complexity | A few high-value data flows, clear contracts |
| Long sales cycle | Named decision-makers, success criteria, conversion terms |
| Building both divisions too early | Thermal Reserve first, Commercial Reserve staged |

**10.5%**

**Contingency reserve**

QAR 195,000 held back. Spent only with documented approval from founders and investors.

○ VISION · NOT BUILT

# One primitive, many industries.

Any asset that deteriorates, cures or loses viability under the stress of time needs this architecture.

### Biopharma ZERO-TRUST COMPLIANCE

| | |
|---|---|
| PHYSICS | stability budgets |
| CALCULATES | active-cooling intercepts on the tarmac |

### Energy THE GRID TRACE

| | |
|---|---|
| PHYSICS | grid signals, inverter telemetry |
| CALCULATES | grid flexibility reserve |

**TRACE → STATE → RESERVE → DECISION**

BUILT FIRST FOR THE FOOD COLD CHAIN

### Space cargo THE ORBITAL TRACE

| | |
|---|---|
| PHYSICS | propellant bleed, thermal cycling |
| CALCULATES | mission reserve for manoeuvres |

### Defence THE KINETIC TRACE

| | |
|---|---|
| PHYSICS | munition shock and degradation |
| CALCULATES | operational readiness reserve |

THERMAL TRACE · BY GENERATIVE GAINS

# Thank you.

GENERATIVE GAINS

**Generative Gains** is raising **QAR 1.85M** to prove Thermal Trace in the field: that what a failing fridge costs can be measured, and recovered.

01

## Najeeb Abdi

MATH · FRONTEND

02

## Islambek

HARDWARE ENGINEER

03

## Param Anand Trimbake

FRONTEND

04

## Ahad Hussain

AI ENGINEER

05

## Robin Thomas

BUSINESS · PRODUCT

**Generative Gains** · Pre-seed · QAR 1.85M · Qatar first, GCC next

github.com/Isl-d/Reboot-The-Earth-Cold-Chain-Logistics